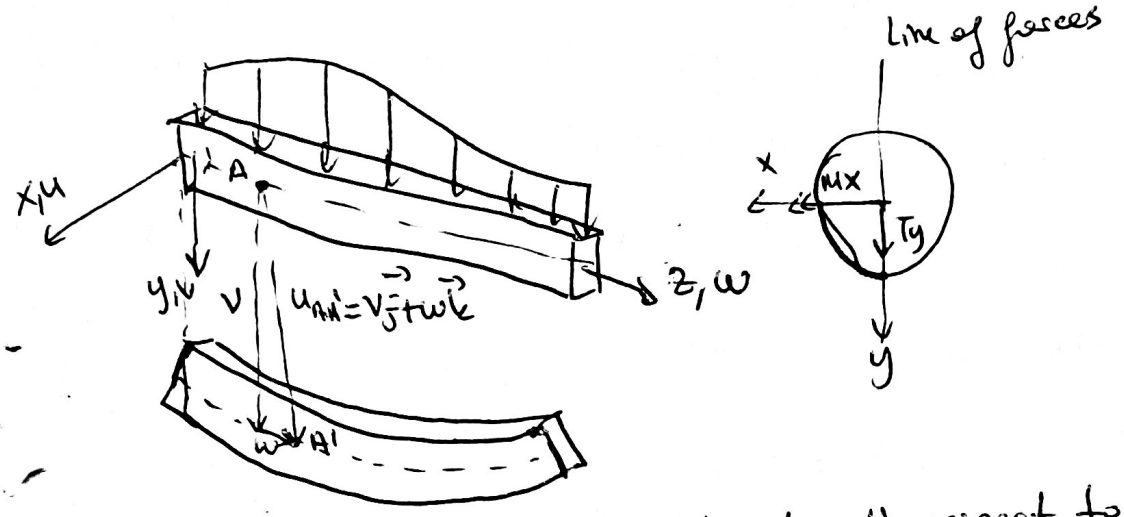
ELASTIC CURVE

Beams under external loadings leaves their initial positions as shown in Figure.



If the cross-section is symmetrical with respect to both the line of forces and bending plane, a point A on the beam axis goes to A' and the displacement vector can be written

$$\vec{u}_{AA'}(z) = v(z)\vec{j} + w(z)\vec{k}$$

In general $w(z) \ll v(z)$ and can be considered as zero. Hence

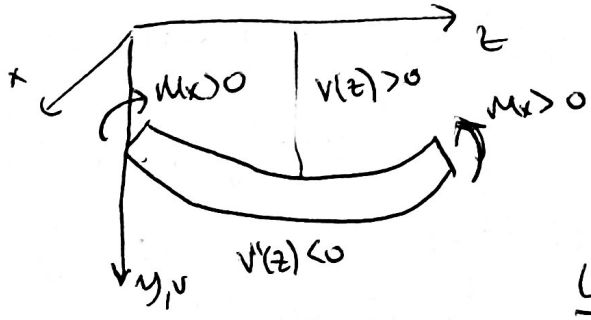
$$\vec{u}(z) = v(z)\vec{j}$$

Differential Equation of the elastic Curve

Recalling the relation between moment and curvature, one can written together with curvature-displacement

$$\left.\begin{aligned} \kappa_x &= \frac{1}{\rho_x} = \frac{M_x}{EI_x} \\ \frac{1}{\rho_x} &= \pm \frac{v''}{[1+(v')^2]^{3/2}} \end{aligned}\right\} \Rightarrow \pm \frac{v''}{[1+(v')^2]^{3/2}} = \frac{M_x}{EI_x} \quad (1)$$

where; $v'' = \frac{d^2v}{dz^2}$; $v' = \frac{dv}{dz}$



Sign convention: While $M_x > 0$, v' is decreasing with increasing values of z Hence $v'' < 0$. So (-) should adopted for moment curvature relation

Linearization: The solution of (1) is called elastica. However, for shallow curves $(v')^2 \ll 1$ denominator is assumed to be 1.

In view of the above simplification, for uniform bending, one can write

$$EI_x\, v''(z) = -M_x \quad (2)$$

*Since (2) is linear, superposition can be applied.

uniform bending: Moment is constant along the beam axis.
In this case elastic curve is a piece of circle

In the case of nonuniform bending, elastic curve takes a general form.

remembering that the $T_y = \frac{dM_x}{dz}$ and $q_y = -\frac{dT_y}{dz}$

$$\left(EI_x v''\right)' = -\frac{dM_x}{dz} = -T_y$$

$$\left(EI_x v''\right)'' = -\frac{dT_y}{dz} = -q_y$$

if EI_x = constant

$$EI_x v''' = -T_y$$

$$EI_x v'''' = q_y \rightarrow \boxed{EI_x \frac{d^4 v}{dz^4} = q_y}$$

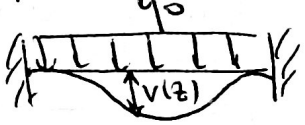
geometric and mechanical quantities for beams

geometric
- $v(z) \rightarrow$ displacement
- $v'(z) = -\Omega_x(z) \rightarrow$ slope or rotation of cross-section

mechanical
- $EI_x v''(z) = -M_x(z) \rightarrow$ Moment (bending)
- $EI_x v'''(z) = -T_y(z) \rightarrow$ shear force
- $EI_x v''''(z) = q_y(z) \rightarrow$ distributed load

In the case of statically indetermined systems:

Integrating diff equation four times ($q_y = q_0$ = constant



$$EIv'''' = q_0 \qquad EI_x v'''(z) = q_0 z + C_1$$

$$EI_x v''(z) = \frac{1}{2} q_0 z^2 + C_1 z + C_2$$

$$EI_x v'(z) = \frac{1}{6} q_0 z^3 + \frac{1}{2} C_1 z^2 + C_2 z + C_3$$

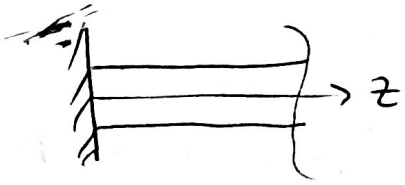
$$EI_x v(z) = \frac{1}{24} q_0 z^4 + \frac{1}{6} C_1 z^3 + \frac{C_2}{2} z^2 + C_3 z + C_4$$

Both boundary and continuity conditions in terms of both geometric and mechanic quantities is required.

Boundary conditions

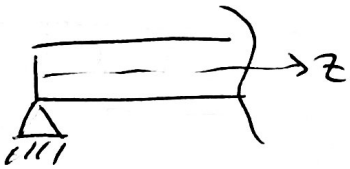
Fixed Support : Both rotation and displacement is constrained

$v = v' = 0$ => geometric boundary conditions



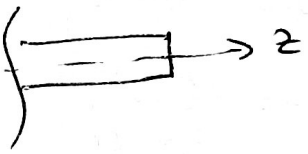
Hinge support : Displacements are zero. If there are no external moment acting on the support moment is also zero

for $z=0$ $v=0$ and $v''=0$



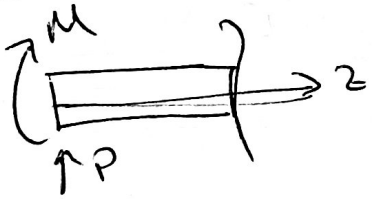
Free End : Shear force and moment are both zero. displacements and rotation is free

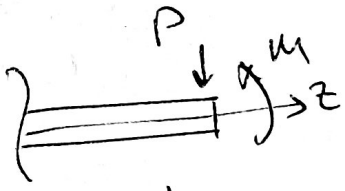
$v'' = v''' = 0$



Free End with external load

or





for $z=0$

$M_x = -EI_x v'' = M$

$T_y = -EI_x v''' = P$

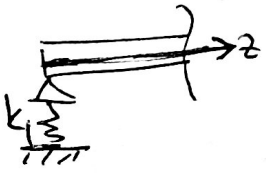
for $z=L$

$M_x = -EI_x v'' = M$

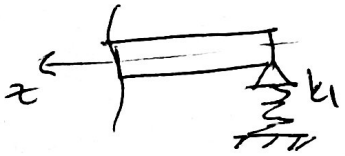
$T_y = -EI_x v''' = P$

Pin support with elastic spring

$-EI_x v''' = k_1 v$ $\quad v'' = 0$



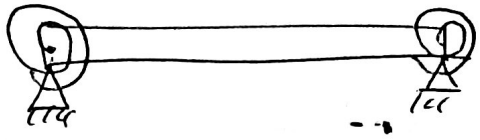
$-EI_x v''' = -k_1 v$ $\quad v'' = 0$



Pin support with rotational spring







for $z=0$

$M_{spr}=M_x=-EI_x v''=-k_2 v'$

$EI_x v''=k_2 v'$

for $z=L$

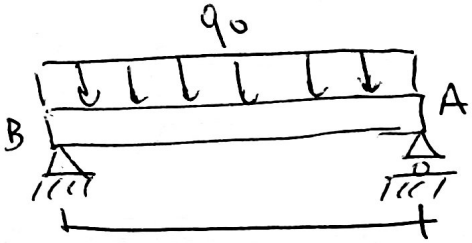
$M_{spr}=M_x=EI_x v''=kv'$

$EI_x v''=-kv'$

$v=0$

$v=0$

Ex:



Find the maximum displacement and rotation at point B under given loading

1. way of solution

Integrating 4th order differential equation

$$EIv^{IV}(z)=q_0 \rightarrow EIv'''=q_0 z+C_1$$

$$EIv''=\frac{1}{2}q_0 z^2+C_1 z+C_2$$

$$EIv'=\frac{1}{6}q_0 z^3+C_1\frac{z^2}{2}+C_2 z+C_3$$

$$EIv=\frac{1}{24}q_0 z^4+C_1\frac{z^3}{6}+C_2\frac{z^2}{2}+C_3 z+C_4$$

Boundary conditions

$v|_{z=0}=0 \Rightarrow C_4=0$

$v''|_{z=0}=0 \Rightarrow C_2=0$

$v''|_{z=L}=0 \quad \frac{1}{2}q_0 L^2+C_1 L=0 \rightarrow C_1=-\frac{q_0 L}{2}$

$v|_{z=L}=0 \quad \frac{1}{24}q_0 L^4+\frac{1}{6}\left(-\frac{1}{2}q_0 L\right)L^3+C_3 L=0 \Rightarrow C_3=\frac{1}{24}q_0 L^3$

substituting boundary conditions into integrated equation

$$EIv = \frac{1}{24} q_0 L^3 z - \frac{1}{12} q L z^3 + \frac{1}{24} q_0 z^4 \quad (0 < z < L)$$

maximum displacement occurs at $v' = 0$

$$EIv' = \frac{1}{24} q L^3 - \frac{1}{4} q_0 L z^2 + \frac{1}{6} q_0 z^3 = 0$$

$$z_1 = 1.37L \quad z_2 = -0.37L \quad z_3 = 0.5L$$

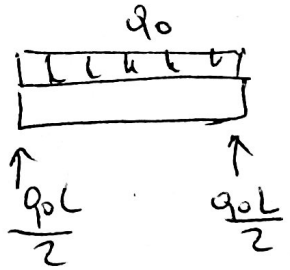
v_{max} occurs at $z_3 = z = 0.5L$

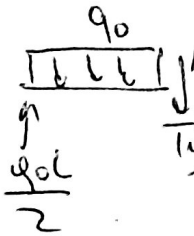
$$v_{max}\Big|_{z=\frac{L}{2}} = \frac{5 q_0 L^4}{384 EI}$$

rotation at B

$$\theta_B = v'(z)\Big|_{z=0} = \frac{q_0 L^3}{24 EI}$$

2nd way of solution: find moment ~~Equation~~





$$M_x = \frac{1}{2} q_0 L z - \frac{1}{2} q_0 z^2$$

using the relation $M = -EIv''$

$$-EIv'' = \frac{1}{2} q_0 L z - \frac{1}{2} q_0 z^3$$

$$-EIv' = \frac{1}{4} q_0 L z^2 - \frac{1}{6} q_0 z^3 + C_1$$

$$-EIv = \frac{1}{12} q_0 L z^3 - \frac{1}{24} q_0 z^4 + C_1 z + C_2$$

Boundary condition

$$v\Big|_{z=0} = 0 \quad C_2 = 0 \qquad v\Big|_{z=L} = 0 \quad C_1 = -\frac{1}{24} q_0 L^3$$

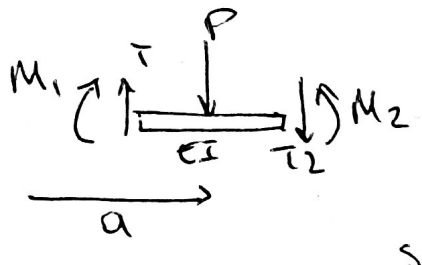
$$EIv = \frac{1}{24} q_0 (L^3 z - 2 L z^3 + z^4)$$ (same as the previous equation)

then find v_{max} and θ_B

Discontinuity in Elastic Curve

If there are single force, couple (Moment) and hinges located at some points on Beams, then, the various derivatives of $v(z)$ includes discontinuities. As a result, it is required that the beam must be seperated into several regions depending on the discontinuities.

Single Force:



$V(z) = V_1(z) \quad z < a$

$V(z) = V_2(z) \quad z > a$

Equilibrium in the vertical direction gives

$$T_2 - T_1 = P$$

differential relations

$$T_1(z) = -EI\, v_1'''(z) \quad \text{and} \quad T_2 = -EI\, v_2'''(z)$$

Substituting in the first equation, one can obtain

$$v_2''' - v_1'''\Big|_{z=a} = \frac{P}{EI} \quad \text{(discontinuity condition)}$$

at $z = a$

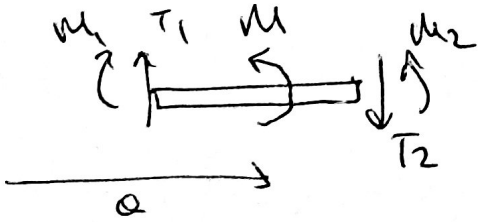
$$v_1\Big|_{z=a} = v_2\Big|_{z=a}$$

compatibility condition

$$v_1'\Big|_{z=a} = v_2'\Big|_{z=a}$$

$$v_1''\Big|_{z=a} = v_2''\Big|_{z=a}$$

Single (Couple) Moment:



One can write $M_1 - M_2 = M$

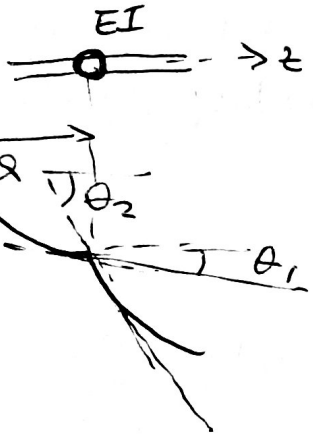
$$M_1(z) = -EI\, v_1''(z) \qquad M_2(z) = -EI\, v_2''(z)$$

discontinuity condition

$$\left(v_2'' - v_1''\right)\Big|_{z=a} = \frac{M}{EI}$$

Compatibility conditions are same as the previous case!

Hinge: Creates a discontinuity in the first derivative of $V(z)$



$\theta = \theta_2 - \theta_1$

$\theta_1 = V_1'$

$\theta_2 = V_2'$

Compatibility conditions

$$V_1 = V_2 \Big|_{z=a}$$

$$V_1''' = V_2''' \Big|_{z=a}$$

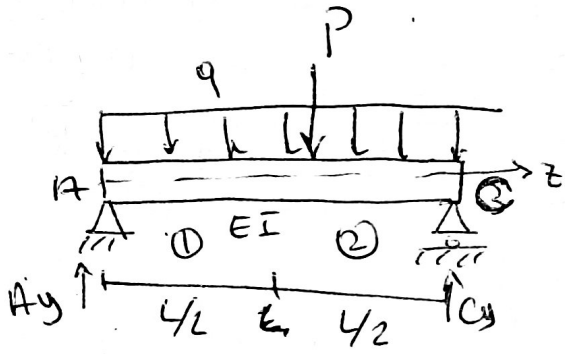
Hinge condition

$$V_1'' \Big|_{z=a} = V_2'' \Big|_{z=a} = 0$$

discontinuity condition

$$V_2' - V_1' \Big|_{z=a} = \theta$$

Ex:

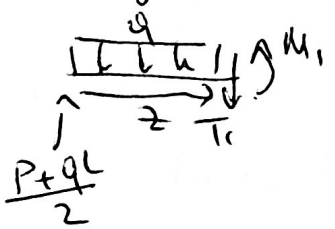


Find the expression that represent the elastic curve. Determine maximum displacement.

Reaction forces;

$$C_y = A_y = \frac{P+qL}{2}$$

① region $0 \le z \le \frac{L}{2}$



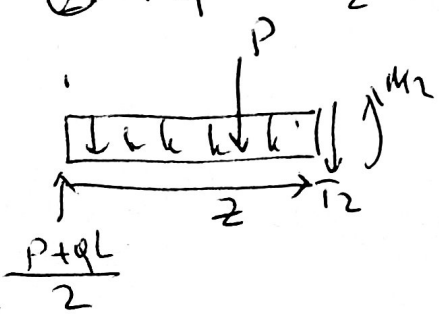
$$M_1 = \frac{P+qL}{2} \cdot z - \frac{1}{2} q z^2$$

$$-EI v_1'' = M_1 \Rightarrow EI v_1'' = -\frac{1}{2}(P+qL)z + \frac{1}{2} q z^2$$

$$EI v_1' = \frac{1}{4}(P+qL)z^2 + \frac{1}{6} q z^3 + C_1$$

$$EI v = -\frac{1}{12}(P+qL)z^3 + \frac{q z^4}{24} + C_1 z + C_2$$

② region $\frac{L}{2} \le z \le L$



$$M_2 = \frac{P+qL}{2} z - \frac{1}{2} q z^2 - P\left(z - \frac{L}{2}\right)$$

$$= \frac{1}{2}\left(LP + (Lq - P)z - q z^2\right)$$

$$EIv'' = -M_2 \Rightarrow EIv_2'' = -\frac{PL}{2} - \frac{qL-P}{2}z + \frac{1}{2}qz^2$$

$$EIv_2' = -\frac{1}{2}PLz - \frac{1}{4}(qL-P)z^2 + \frac{1}{6}qz^3 + C_3$$

$$EIv_2 = -\frac{PL}{4}z^2 - \frac{1}{12}(qL-P)z^3 + \frac{qz^4}{24} + C_3z + C_4$$

$$v_1\Big|_{z=0} = 0 \rightarrow C_2 = 0$$

$$v_2\Big|_{z=L} = 0 \rightarrow C_4 + C_3L - \frac{PL^3}{6} - \frac{qL^4}{24} = 0$$

$$v_1\Big|_{z=\frac{L}{2}} = v_2\Big|_{z=\frac{L}{2}} \rightarrow \frac{C_1L}{2} + \frac{1}{24}PL^3 = C_4 + \frac{C_3L}{2}$$

$$v_1'\Big|_{z=\frac{L}{2}} = v_2'\Big|_{z=\frac{L}{2}} \rightarrow C_1 + \frac{PL^2}{8} = C_3$$

Solve for C_1, C_2, C_3 and C_4

$$C_1 = \frac{3PL^2 + 2qL^3}{48} \qquad C_2 = 0$$

$$C_3 = \frac{9PL^2 + 2qL^3}{48} \qquad C_4 = \frac{-PL^3}{48}$$

$$EIv_1 = \left(\frac{PL^2}{16} + \frac{qL^3}{24}\right)z - \frac{(P+qL)}{12}z^3 + \frac{qz^4}{24} \qquad 0 < z < \frac{L}{2}$$

$$EIv_2 = -\frac{PL^3}{48} + \left(\frac{3PL^2}{16} + \frac{qL^3}{24}\right)z - \frac{PL}{4}z^2 - \frac{(qL-P)}{12}z^3 + \frac{qz^4}{24} \qquad \frac{L}{2} < z < L$$

$$v_{max} = v_1\Big|_{z=\frac{L}{2}} = \frac{L^3}{384EI}(8P + 5qL)$$

1.4.2. Mohr method/moment-area method

Moment graph of the beam must be known for using this method. This method calculates the elastic curve by using a mechanical analogy. This analogy has two group of variations.

First group: From the equilibrium conditions on a beam, the following equations can be written between distributed force=q, shear force= T and bending moment=M:

$$\frac{dT}{dz} = -q \qquad \frac{dM}{dz} = T \qquad \frac{dT}{dz} = \frac{d^2M}{dz^2} = -q$$

Second group: A beam has also the following relations from definition and geometric basis between

$curvature = \frac{1}{\rho x}$ $slope = \theta$ and $displacement$ v

$$\frac{d\theta}{dz} = \frac{d^2v}{dz^2} = -\frac{1}{\rho_x} = -\frac{M_x}{EI_x}$$

$$\frac{d\theta}{dz} = -\frac{M_x}{EI_x} \qquad \frac{dv}{dz} = \theta \qquad \frac{d^2v}{dz^2} = -\frac{M_x}{EI_x}$$

It can be seen from the comparison of the sets of equations that there are mathematical analogy pairs between the six quantities:

Force	$\Leftrightarrow$ Curvature	$q \Leftrightarrow -\frac{1}{\rho_x}$
Shear force	$\Leftrightarrow$ Slope	$T \Leftrightarrow \theta$
Bending moment	$\Leftrightarrow$ Displacement	$M_x \Leftrightarrow v$

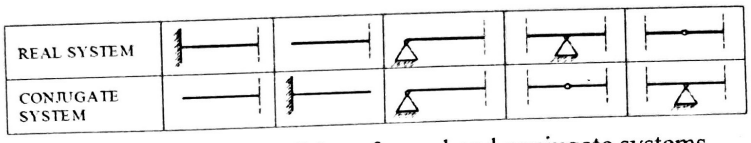
By assuming the curvature of the real beam as a force affecting on the conjugate beam, the shear force diagram of the conjugate beam gives the rotation of the real beam, the moment diagram of the conjugate beam gives the displacement diagram of the real beam.

Force ⟺ Curvature

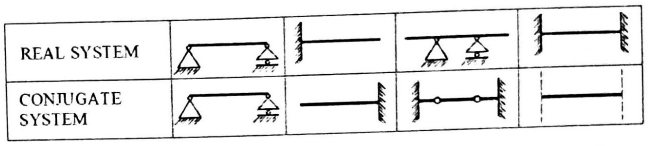
Shear force ⟺ Slope

Bending moment ⟺ Displacement

Tableo 1.1 a) Boundary conditions for real and conjugate systems



Tablo 1.1 b) Boundary conditions for real and conjugate systems



Based on this basic idea, Mohr's method help us find the slopes and displacements of the beam by means of shear force and bending moment diagrams of another beam loaded by fictitious curvature load. This fictitious beam is called conjugate beam system of the real beam system.

The conjugate beam end conditions may not be the same as the end conditions of the actual system. Appropriate end conditions must be determined by considering the analogies between the above quantities.

Deflection and slope are zero at the fixed support on real beam; conjugate beam moment and shear forces must be zero in this section. Hence, fixed support on real system is replaced with free end on conjugate system.

By utilizing the reciprocity in analogy, free end of the actual beam system is replaced with fixed end of the conjugated beam system.

If the support is an immovable hinge in the vertical direction on the real system, then although $v = 0$, the rotation angle θ is not zero at this support. So, on the conjugate system, this support have to be a hinge without deflection, because such a support has zero moment (means zero displacement in the real system) and the shear force is not zero (means rotation angle is different than zero).

An intermediate support on real system corresponds to a hinged joint on the conjugate system.

A hinged joint on real system corresponds to an intermediate support without deflection on the conjugate system.

After this preliminary information, the solution procedure of the method to be followed is given below:

1) First the moment diagram Mx is drawn for the real system. Then the values of moment diagram are divided by $E I_x$ to obtain the curvature diagram.

2) Conjugate of the real system is obtained. The curvature diagram of the real systems is considered as the load diagram for the conjugate system.

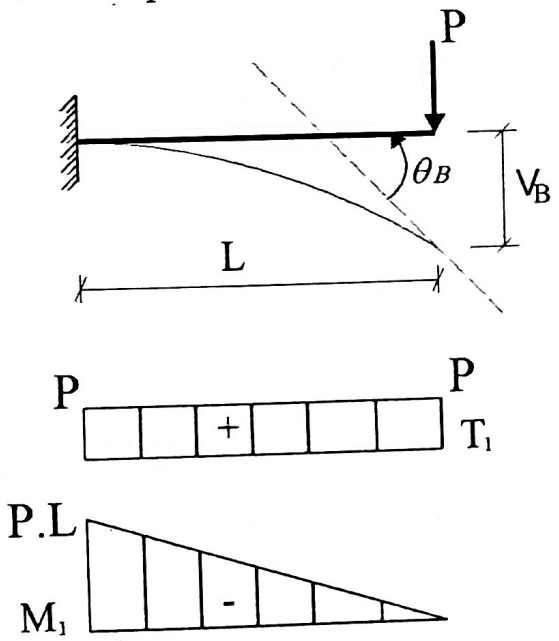
3) The shear force diagram of the conjugate system gives the slope diagram of the real system.

4) Bending moment diagram of the conjugate system gives the displacements of the actual system.

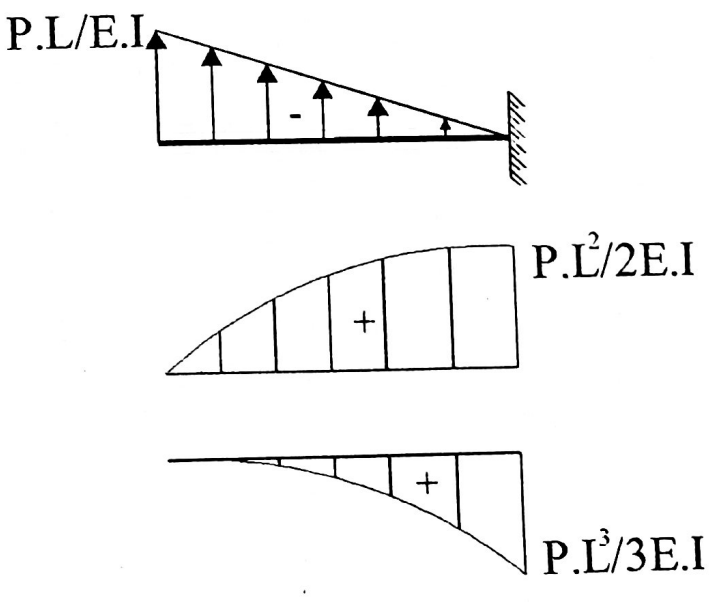
This method is particularly useful for obtaining rotations and displacements of the beams loaded with concentrated forces and moments.

Problem 1.6: In the figure shown below, a cantilever beam is loaded with a concentrated force P. Find the elastic curve of the beam, calculate the angle of rotation and displacement at the free end of the beam.

Obtained quantities for real system:

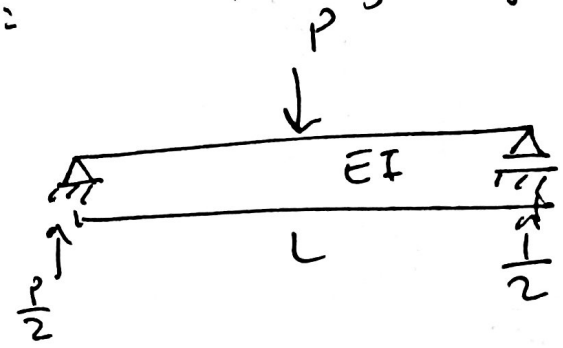


Obtained quantities for conjugate system:



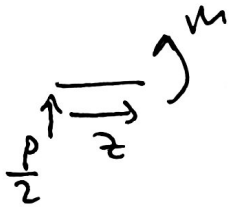
Solve using Integration and Mohr Method

Ex:

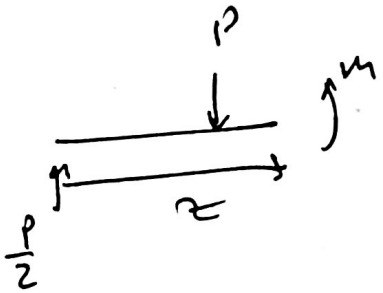


$$V_1 = \frac{-Pz^3}{12EI} + \frac{PL^2}{16EI}\cdot z$$

$$V_{max} = V_1\Big|_{z=\frac{L}{2}} = \frac{2PL^2}{96EI} = \frac{PL^3}{48EI}$$



$$M_1 = \frac{Pz}{2}$$



$$M_2 = \frac{Pz}{2} - Pz + \frac{PL}{2}$$

$$= -\frac{Pz}{2} + \frac{PL}{2}$$

$$EIv_1'' = -\frac{Pz}{2}$$

$$EIv_1' = -\frac{Pz^2}{4} + C_1$$

$$EIv_1 = -\frac{Pz^3}{12} + C_1 z + C_2$$

$$EIv'' = \frac{Pz}{2} - \frac{PL}{2}$$

$$EIv' = \frac{Pz^2}{4} - \frac{PL}{2}z + C_3$$

$$EIv = \frac{Pz^3}{12} - \frac{PL}{4}z^2 + C_3 z + C_4$$

(1) $v_1\Big|_{z=0} = 0 \qquad C_2 = 0$

(2) $v_2\Big|_{z=L} = 0 \qquad \frac{PL^3}{12} - \frac{PL^3}{4} + C_3 L + C_4 = 0 \Rightarrow C_3 = \frac{9PL^2}{48}$

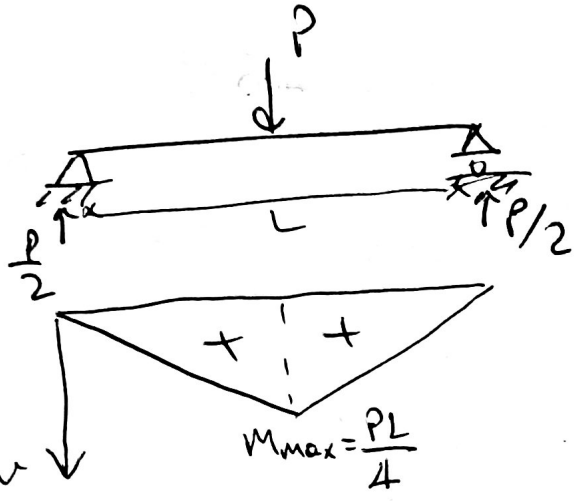
(3) $v_1\Big|_{z=\frac{L}{2}} = v_2\Big|_{z=\frac{L}{2}} \qquad \frac{-PL^3}{96} + C_1\frac{L}{2} = \frac{PL^3}{96} - \frac{PL^3}{16} + C_3\frac{L}{2} + C_4$

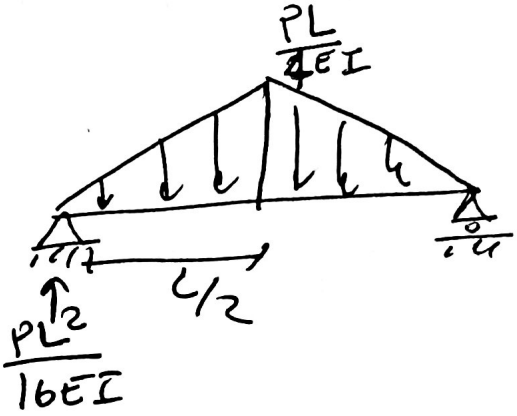
(4) $v_1' = v_2'\Big|_{L/2} \qquad \frac{-PL^2}{16} + C_1 = \frac{PL^2}{16} - \frac{PL^2}{4} + C_3$

$$C_1 = -\frac{PL^2}{8} + C_3 \Rightarrow C_1 = \frac{3PL^2}{48}$$

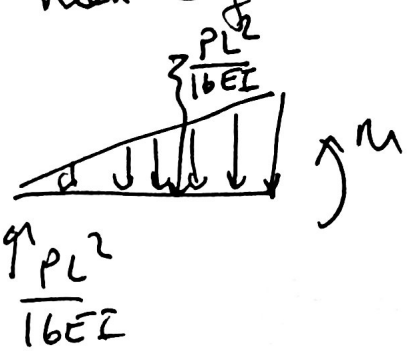
(3)'den $\quad -\frac{PL^3}{96} + \frac{PL^3}{16} + C_3\frac{L}{2} = \frac{PL^3}{96} - \frac{PL^3}{16} + C_3\frac{L}{2} + C_4 \qquad C_4 = -\frac{5PL^3}{48}$

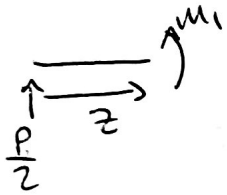
Grou... ...urname...



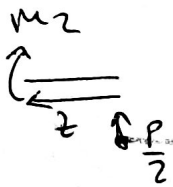


max deflection





$M_1 = \frac{P}{2} z$

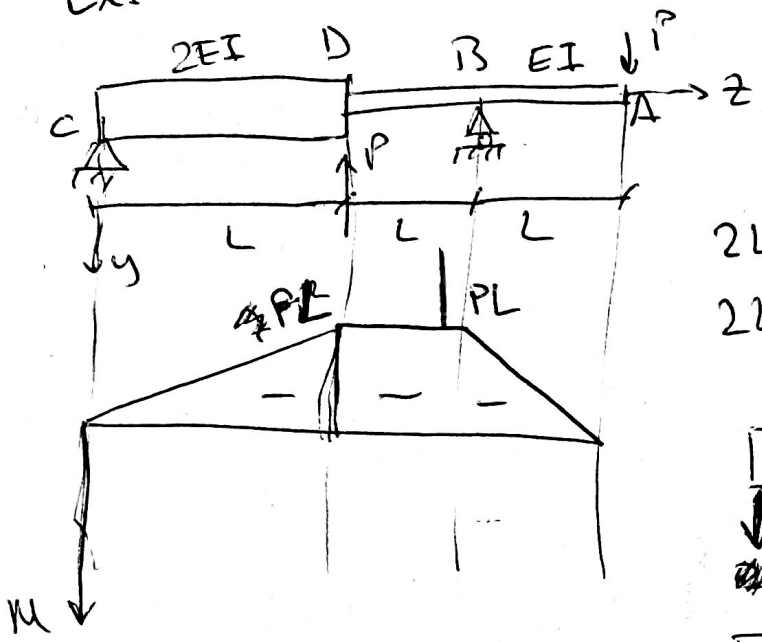


$M_2 = \frac{P}{2} \cdot z$

$$M = \frac{PL^2}{16EI} \cdot \frac{L}{2} - \frac{PL^2}{16EI} \cdot \frac{L}{2} \cdot \frac{1}{3}$$

$$= \frac{PL^3}{48EI}$$

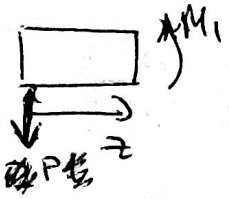
Ex:



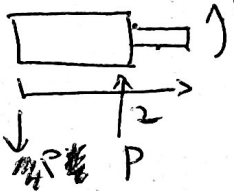
Find the displacement v_A (at A) and θ_B (at point B) using Mohr method.

$$2L \cdot C_y = -PL - PL = -2PL = -2P = -1$$

$$2L \cdot B_y = 2PL \Rightarrow B_y = P$$

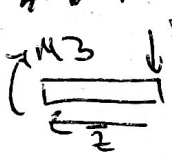


$$M_1 = -Pz$$



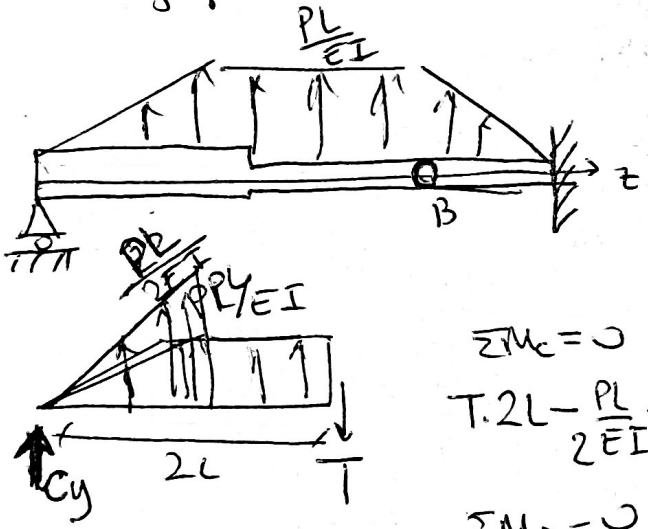
$$M_2 = P(z-L)$$

$$= -PL$$



$$M_3 = -Pz$$

Conjugate Beam

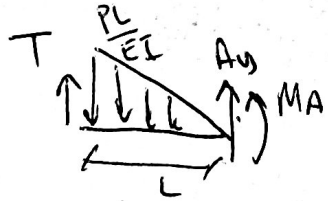


$$\Sigma M_C = 0$$

$$T \cdot 2L - \frac{PL}{2EI} \cdot L \left(\frac{2L}{3}\right) - \frac{PL}{EI} \cdot L \left(\frac{L}{2} + L\right) = 0 \qquad T = \theta_B = \frac{5PL^2}{6EI}$$

$$\Sigma M_B = 0$$

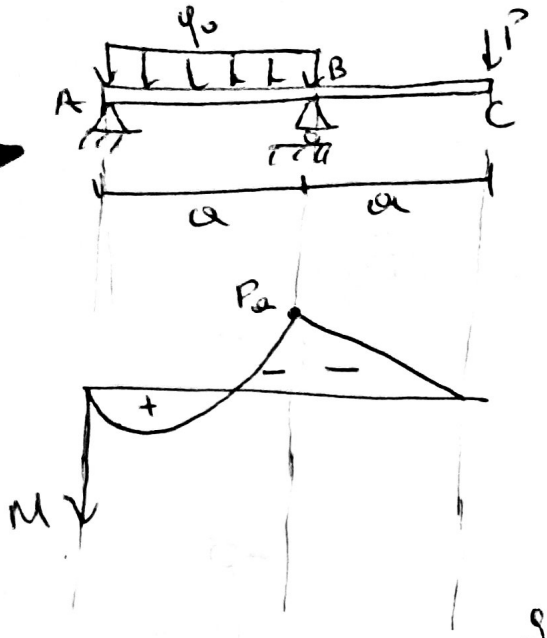
$$C_y \cdot 2L + \frac{PL}{2EI} \cdot \frac{L}{2} \cdot \left(\frac{L}{3} + L\right) + \frac{PL}{EI} \cdot L \cdot \frac{L}{2} = 0$$



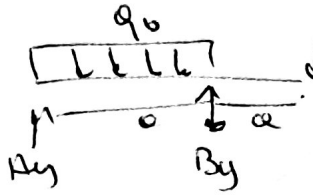
$$\Sigma M_A = 0 \qquad M_A - T \cdot L + \frac{PL}{EI} \cdot \frac{L}{2} \cdot \frac{2L}{3} = 0$$

$$M_A = v_A = \frac{7PL^3}{6EI}$$

Ex:



Find the displacement and rotation at point C using conjugate beam method (Mohr)

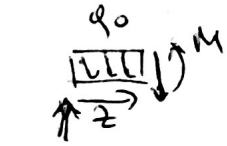


$$-By \cdot a + P \cdot 2a + q_0 \cdot a \cdot \frac{a}{2} = 0$$

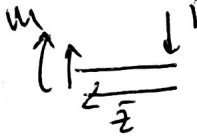
$$By = 2P + \frac{q_0 a}{2}$$

$$Ay \cdot a + P \cdot a - q_0 \cdot a \cdot \frac{a}{2} = 0$$

$$Ay = \frac{q_0 a}{2} - P$$

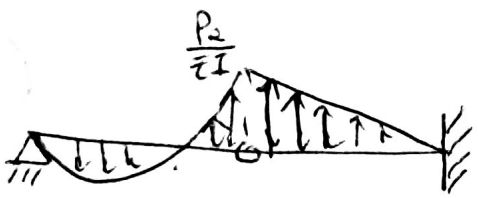


$$M = \left(\frac{q_0 a}{2} - P\right) \cdot z - q_0 \frac{z^2}{2} \quad 0 \le z \le a$$

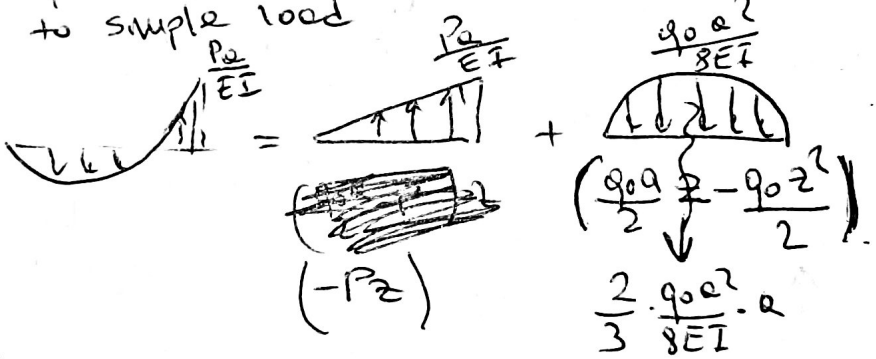


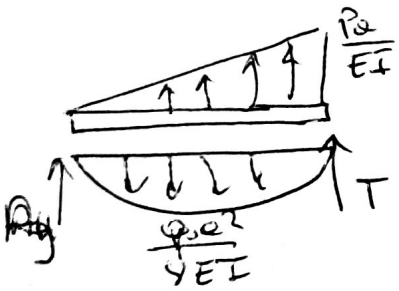
$$M = -P z \quad a < z < 0$$

Conjugate beam



Parabolic load can be seperated into to simple load

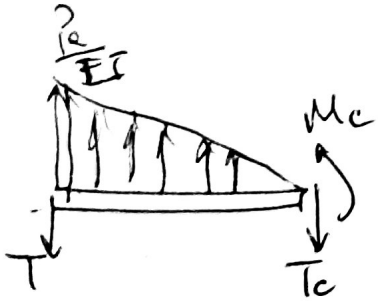




$$\sum M_A = 0$$

$$T \cdot a + \frac{1}{2} \frac{Pa^2}{EI} \cdot \frac{2}{3} a - \frac{q_0 a^2}{8EI} \cdot \frac{2}{3} a \cdot \frac{a}{2} = 0$$

$$T = \frac{q_0 a^3}{24EI} - \frac{Pa^2}{3EI}$$



$$\theta_C = T_C = \frac{Pa^2}{2EI} - \left(\frac{q_0 a^3}{24EI} - \frac{Pa^2}{3EI}\right) = \frac{5Pa^2}{6EI} - \frac{q_0 a^3}{24EI}$$

$$y_C = M_C = \frac{Pa^2}{EI} \cdot \frac{2}{3} a - \left(\frac{q_0 a^3}{24EI} - \frac{Pa^2}{3EI}\right) a$$

$$= \frac{2Pa^3}{3EI} - \frac{q_0 a^4}{24EI}$$